Visualizing Temperatures in Dublin

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Introduction

This is my report for the Data Programming CA2. For my topic I have chosen to visualize and analyze temperature data for Dublin. For that purpose, I chose 6 weather stations in Dublin to create a dataset from them to represent Dublin in its entirety.

Installation

The required libraries are listed in the README.md which comes included in the project zip.

All other required files should be included in the zip.

Project Objectives

The objectives I had when starting this project were to visualize trends in the temperature measurements from all over Ireland. I had found lots of data from all around the country and was ready to get to work. Soon after starting to plan out the project, I realized it was too ambitious and had to reduce my scope significantly.

I started looking for datasets from Dublin weather stations instead. There weren't many that were still operating but I found enough data that would be suitable. The main station would be the Dublin Airport weather station, which has a dataset on data.gov.ie starting in 1941.

This dataset was complimented by multiple other stations around Dublin that weren't as old. These datasets would help ensure the data was an accurate representation of Dublin as a whole.

My objective was to find upward trends in the temperature data, which would be evident as the data I collected covered around 80 years of Dublin's temperature.

Dataset Introduction

The datasets used were retrieved from https://data.gov.ie/dataset. Six different datasets were chosen representing six weather stations in Dublin. All six datasets track mean 10cm soil temperature as well as 09UTC minimum grass temperature (gmin).

Three of the datasets recorded air temperature while the other 3 recorded temperatures. Temperature is measured in the open air and includes factors that affect how hot it feels to humans. Air temperature is measured in a box to minimize the effect of other factors on the measurement.

The grass minimum temperature is measured by a minimum thermometer in the open air with its bulb at the top of the grass. It is usually used along with the next value to help farmers decide when to plant their crops.

Mean 10cm soil temperature is average temperature of soil at a depth of 10cm. It is a very useful measurement as unlike the air (which fluctuates temperature rapidly) the soil maintains its temperature for some time. It is also a very useful measurement for farmers as plants need to be able to operate at this depth.

With all these values you can get a general sense of the temperature of the entirety of Dublin, not just the air temperature.

Pre-Processing Tasks

The first task was to re-order the date from each entry into a field more easily manipulated by python and pandas. The date goes from “30-Nov-25” to “2025-11-30”, which can easily be transferred to pythons datetime. It is easy to store it as a string and then convert it to a datetime object when required in the insight.py files. From the datetime format it can be turned into any format required.

Multiple tables also contained duplicate entries. This was corrected by putting each date into a variable and checking it whenever a new line in the csv file was passed over. If the date is the same as the previous line then nothing is calculated or inserted from that row, as it was empty anyway.

The datasets also contained a lot of empty values. These (if left unchecked) would cause a lot of problems with the derivation of new values. So, each entry is checked to see if it is a string with a space and if so, it is replaced by None. This ensures that the database is not filled with empty strings that make analyzing the data impossible. If the value is not an empty string then it is set to its value from the csv file and added to the row.

The min and max are checked and if both are present then new values are made. These values are the mean temperature and the difference in max and min temperature. These values are derived from the min and max temp value in each table as it is entered to the database in case individual stations are viewed on their own. These values are also calculated for the Dublin table.

Finally, the 6 tables are loaded from the database, and then each row is added to a DataFrame. Since all 6 datasets begin at different dates they must be indexed by the date and inserted, so the dates remain consistent.

Each table is loaded and each value is put into their own DataFrame. After all the tables have been deconstructed like this the DataFrame’s are aggregated to coalesce their data into one DataFrame for Dublin.

The mean and difference air temperature and temperature are calculated and added as well. The final Dublin table contains 10 fields (not counting the date). This is assembled in a DataFrame and

then sent to the database by using the to_sql function. No schema is defined for the Dublin table as all the values that were used to create this table were already validated and processed.

Database Storage

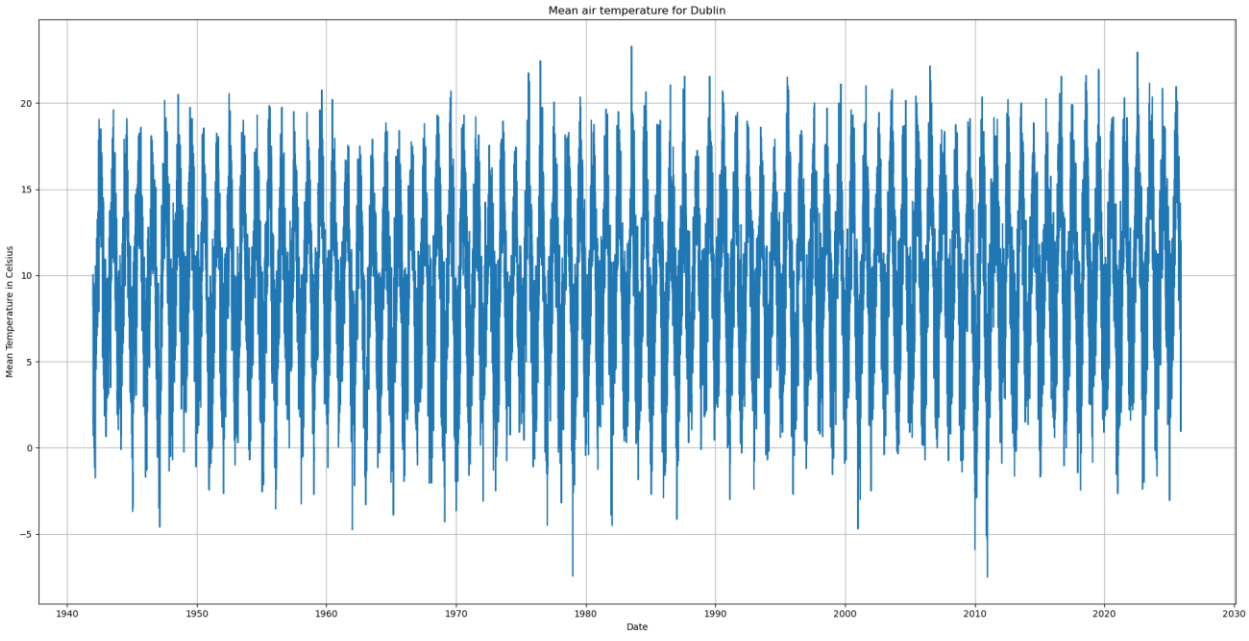
The DataFrames made from the csv files are parsed line by line and inserted into the database into a table of the same name. The schema is very simple and is just a string for the date and all other entries are floats. No other types of data are required so it's best to keep it simple.

For the Dublin database that is created, the DataFrame is uploaded to the database using the pandas to_sql function. This forgoes the need for a schema and since it is only being written to once, a schema is not required.

Analysis

Insight 1

My first insight is a slope I calculated from the mean air temperature of the Dublin table. It was calculated using linear regression, on the date column (changed to UNIX timestamp) and the mean air temperature column. Unfortunately, I was unable to remove the seasonality from the data and so the other values gotten from the regression are inaccurate. Below you can see a graph of the mean air temperature.

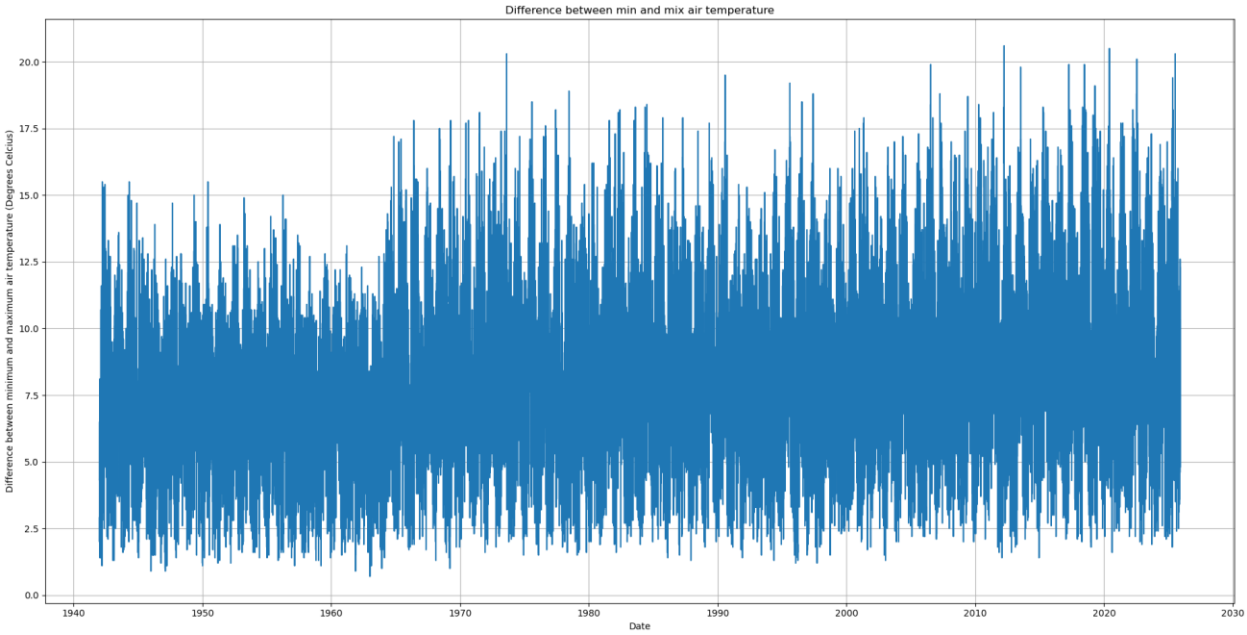


The slope gotten was 3.21e-10, but this was gotten by using UNIX timestamps. Multiplied by the seconds in a year we get the slope for each year, 0.0101. This slope indicates an increase in the mean air temperature of about a hundredth of a degree a year, a tenth of a degree every decade.

```
Slope for seconds: 3.213975158172309e-10
Slope for years: 0.010135592058812194
```

Insight 2

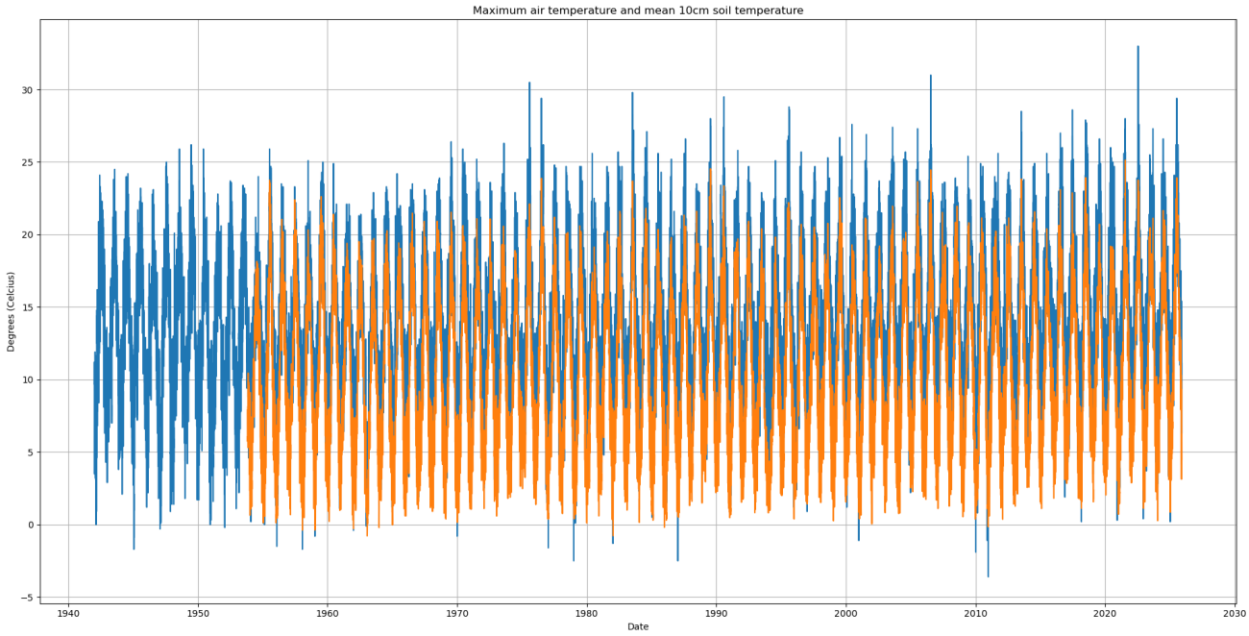
The second insight is the increase in the difference between the daily maximum and minimum recorded temperature. This is called the “diurnal range of temperature” and it represents the daily fluctuation of temperature. Below you can see a graph of the “diurnal range”.



There is likely not enough data prior to 1964 to aggregate into a reliable figure, but the increase can be seen in the data after. This increase means that there will be higher temperature variations and “wilder” weather each day.

Insight 3

For the third insight, I wanted to show the correlation between the soil temperature and max air temperature. This relationship is closely correlated as the soil can absorb heat quickly but takes a lot longer to lose that heat. Below you can see the maximum air temperature (blue) and the mean 10cm soil temperature (orange).



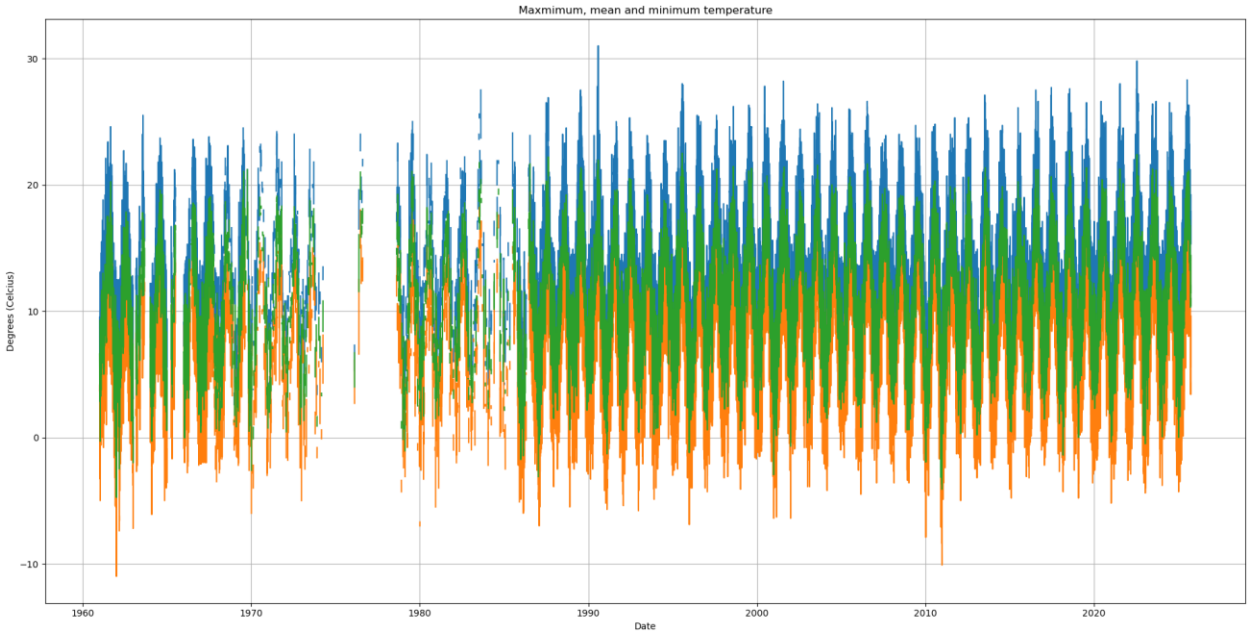
The soil temperature closely mirrors the maximum temperature but is still off as it still takes time to heat up. The correlation coefficients have been calculated using pandas and can be seen below.

	SOIL	MAXTP	MINTP
SOIL	1.000000	0.936437	0.817284

The max temperature has a high coefficient, signaling that they are quite similar. The minimum temperature is still relatively high coefficient but is significantly lower due to the soil retaining heat and being closer to the max temperature.

Insight 4

The last insight is into the increase of “feels-like” temperature. Unlike the air temperature shown before, it is calculated from the air temperature. It also includes factors that affect how the weather feels to people, such as sunshine and the wind. Below you can see the graph of maximum (blue), mean (green) and minimum (orange) temperature.



The increase in max temperatures on this graph is a lot easier to see when compared to the first insight. This is because the other factors make this temperature measurement a lot more volatile, and more relevant to the average person.

Critical Discussion and Challenges

I encountered lots of challenges when completing this project. The biggest being the time limit, I was overconfident in my ability and did not give myself enough time to complete this project. So, in the end I had to rush and make compromises, ideally, I would’ve had a lot more time and I will be very careful to delay starting on my next project.

Another big issue I had with the project was the statistical analysis. I did not have enough time to learn enough to properly analyze a lot of my data. I would’ve liked to learn how to properly remove the seasonality from my data and calculate a proper p-value for that insight.

There are some limitations in my data collected, as you can see in insight 4, there is a large part missing from the graph. The temperature data I found was not all the same time scale and so had to be collected and the mean calculated from each station. This means the later the data the more reliable it is, as multiple stations are likely to have contributed to the value.

In the beginning I had a lot of trouble properly manipulating data with pandas. Especially manipulating columns and rows inside of a DataFrame, and the transfer of data to and from the database confused me a lot. I can confidently say that I have learned a lot about pandas since starting this project and I now realize its usefulness. If I were to make my own project like this in the future, I would utilize pandas for sure.

I was also unable to utilize some of the fields I collected, although I'm sure they contain useful data. The 09 UTC grass minimum temperature was overshadowed by the soil temperature and the extra calculated values for mean and difference of temperature were never used.

Conclusion

I was able to achieve my objectives for the most part, if I had continued with my original objective I would not have gotten the project done in time. Thankfully I reduced my scope significantly and was able to finish before the deadline.

I have gained a lot of experience with pandas, matplotlib and sqlite3. I had used pandas and matplotlib to a limited extent but I had not used sqlite3 before and found the SQL strings confusing to enter.

I was very bad at data manipulation when I started this project, but I can say that I have most definitely improved significantly. I hope I get the chance to use the skills I have gained in a future personal or college project.

Bibliography

https://www.met.ie/climate/

https://www.piarc.org/en/activities/Road-Dictionary-Terminology-Road-Transport/term-sheet/96249-en-grass%20minimum%20temperature#:~:text=Definition%20:%20Temperature%20shown%20by%20a,the%20tops%20of%20grass%20blades

https://docs.scipy.org/doc

https://matplotlib.org/stable/users/getting_started/

https://docs.python.org/3/library/sqlite3.html

Dataset Links

https://data.gov.ie/dataset/dublin-airport-monthly-data

https://data.gov.ie/dataset/dublin-glasnevin-climate-data

https://data.gov.ie/dataset/dublin-merrion-square-climate-data

https://data.gov.ie/dataset/casement-monthly-data

https://data.gov.ie/dataset/dun-laoghaire-climate-data

https://data.gov.ie/dataset/phoenix-park-monthly-data

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